

From Bottleneck to Breakthrough: AI in Chip Verification

How AI is transforming chip design with smarter verification methods

In the world of electronics, integrated circuits (IC) chips are the unseen powerhouse behind progress. Every leap—whether it's smarter phones, more capable cars, or breakthroughs in healthcare and science—relies on chips that are more complex, faster, and packed with more features than ever before. But creating these chips is not just a question of sheer engineering talent or ambition. The design process itself has reached staggering levels of complexity, and with it, the challenge to keep productivity and quality moving forward.

As we push against the boundaries of physics, chipmakers face more than just technical hurdles. The workforce challenges, tight timelines, and the requirements for building reliable chips are stricter than ever. Enormous effort goes into making sure chip layouts follow detailed constraints—such as maintaining minimum feature sizes for transistors and wires, keeping proper spacing between different layers like metal, polysilicon, and active areas, and ensuring vias overlap correctly to create solid electrical connections. These design rules multiply with every new technology generation. For every innovation, there's pressure to deliver more with less. So, the question becomes: How do we help designers meet these demands, and how can technology help us handle the complexity without compromising on quality?

Shifting the paradigm: the rise of AI in electronic design automation

A major wave of change is moving through the entire field of electronic design automation (EDA), the specialized area of software and tools that chipmakers use to design, analyze, and verify the complex integrated circuits inside today's chips. Artificial intelligence is already touching many parts of the chip design flow—helping with placement and routing, predicting

yield outcomes, tuning analog circuits, automating simulation, and even guiding early architecture planning. Rather than simply speeding up old steps, AI is opening doors to new ways of thinking and working.

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Instead of brute-force computation or countless lines of custom code, AI uses advanced algorithms to spot patterns, organize massive datasets, and highlight issues that might otherwise take weeks of manual work to uncover. For example, generative AI can help designers ask questions and get answers in natural language, streamlining routine tasks. Machine learning models can help predict defect hotspots or prioritize risky areas long before sending a chip to be manufactured.

This growing partnership between human expertise and machine intelligence is paving the way for what some call a “shift left” or concurrent build revolution—finding and fixing problems much earlier in the design process, before they grow into expensive setbacks. For chipmakers, this means higher quality and faster time to market. For designers, it means a chance to focus on innovation rather than chasing bugs.

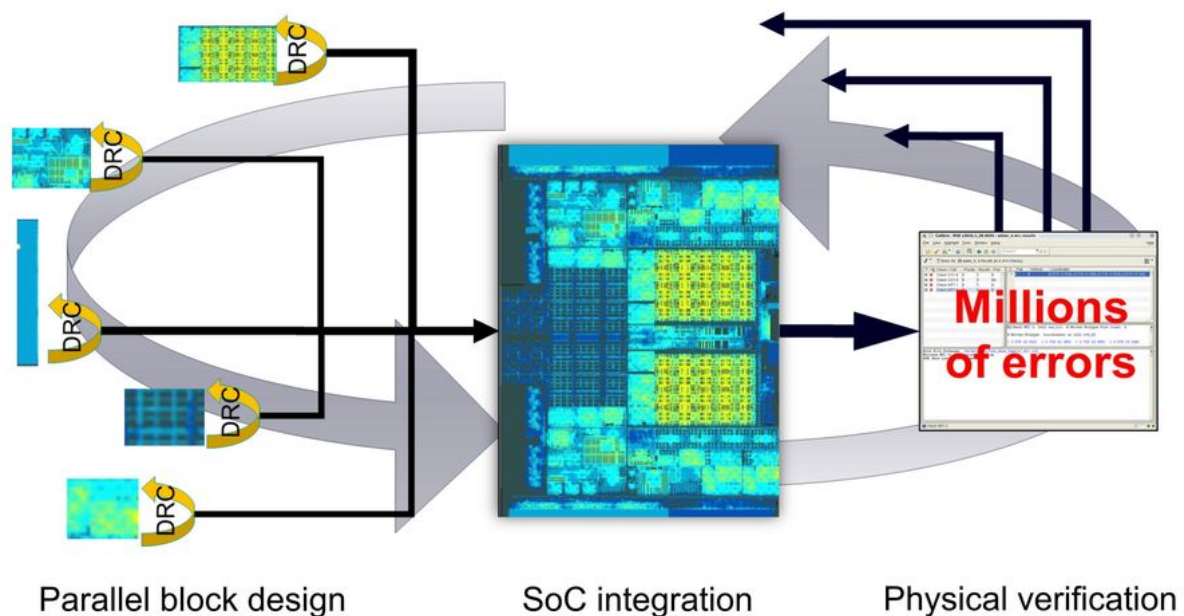


Figure 1. Shift-left and concurrent build of IC chips performs multiple tasks simultaneously that use to be done sequentially.

The physical verification bottleneck: why design rule checking is harder than ever

As chips grow more complex, the part of the design called physical verification becomes a critical bottleneck. Physical verification checks whether a chip layout meets the manufacturer's strict rules and faithfully matches the original functional schematic. Its main goal is to ensure the design can be reliably manufactured into a working chip, free of physical defects that might cause failures later on.

Design rule checking (DRC) is the backbone of physical verification. DRC software scans every corner of a chip's layout for violations—features that might cause defects, reduce yield, or simply make the design un-manufacturable. But today's chips aren't just bigger; they're more intricate, woven from many layers of logic, memory, and analog components, sometimes stacked in three dimensions. The rules aren't simple either. They may depend on the geometry, the context, the manufacturing process and even the interactions between distant layout features.

Traditionally, DRC is performed late in the flow, when all components are assembled into the final chip layout. At this stage, it's common to uncover millions of violations—and fixing these late-stage issues requires extensive effort, leading to costly delays.

To minimize this burden, there's a growing focus on shifting DRC earlier in the flow—a strategy called “shift-left.” Instead of waiting until the entire design is complete, engineers try to identify and address DRC errors much sooner at block and cell levels. This concurrent design and verification approach allows the bulk of errors to be caught when fixes are faster and less disruptive.

However, running DRC earlier in the flow on a full chip when the blocks are not DRC clean produces results datasets of breathtaking scale—often tens of millions to billions of “errors,” warnings, or flags because the unfinished chip design is “dirty” compared to a chip that's been through the full design process. Navigating these “dirty” results is a challenge all on its own. Designers must prioritize which issues to tackle, identify patterns that point to systematic problems, and decide what truly matters. In many cases, this work is slow and

“manual,” depending on the ability of engineers to sort through data, filter what matters, and share findings across teams.

To cope, design teams have crafted ways to limit the flood of information. They might cap the number of errors per rule, or use informal shortcuts—passing databases or screenshots by email to team members, sharing filters in chat messages, and relying on experts to know where to look. Yet this approach is not sustainable. It risks missing major, chip-wide issues that can cascade through the final product. It slows down response and makes collaboration labor-intensive.

With ongoing workforce challenges and the surging complexity of modern chips, the need for smarter, more automated DRC analysis becomes urgent. So what could a better solution look like—and how can AI help bridge the gap?

The rise of AI-powered DRC analysis

Recent breakthroughs in AI have changed the game for DRC analysis in ways that were unthinkable even a few years ago. Rather than scanning line by line or check by check, AI-powered systems can process billions of errors, cluster them into meaningful groups, and help designers find the root causes much faster. These tools use techniques from computer vision, advanced machine learning, and big data analytics to turn what once seemed like an impossible pile of information into a roadmap for action.

AI’s ability to organize chaotic datasets—finding systematic problems hidden across multiple rules or regions—helps catch risks that basic filtering might miss. By grouping related errors and highlighting hot spots, designers can see the big picture and focus their time where it counts. AI-based clustering algorithms reliably transform weeks of manual investigation into minutes of guided analysis.

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Another benefit: collaboration. By treating results as shared, living datasets—rather than static tables—modern tools let teams assign owners, annotate findings and pass exact analysis views between block and partition engineers, even across organizational boundaries.

Dynamic bookmarks and shared UI states cut down on confusion and rework. Instead of “back and forth,” teams move forward together.

Many of these innovations tease at what’s possible when AI is built into the heart of the verification flow. Not only do they help designers analyze the results; they help everyone reason about the data, summarize findings and make better design decisions all the way to tape out.

A real-world breakthrough in DRC analysis and collaboration: Siemens’ Calibre Vision AI

One of the most striking examples of AI-powered DRC analysis comes from Siemens, whose Calibre Vision AI platform is setting new standards for how full-chip verification happens. Building on years of experience in physical verification, Siemens realized that breaking bottlenecks required not only smarter algorithms but rethinking how teams work together and how data moves across the flow.

Vision AI is designed for speed and scalability. It uses a compact error database and a multi-threaded engine to load millions—or even billions—of errors in minutes, visualizing them so engineers see clusters and hot spots across the entire die. Instead of a wall of error codes or isolated rule violations, the tool presents a heat map of the layout, highlighting areas with the highest concentration of issues. By enabling or disabling layers (layout, markers, heat map) and adjusting layer opacity, users get a clear, customizable view of what’s happening—and where to look next.

Using advanced machine learning algorithms, Vision AI analyzes every error to find groups with common failure causes.

But the real magic is in AI-guided clustering. Using advanced machine learning algorithms, Vision AI analyzes every error to find groups with common failure causes. This means designers can attack the root cause once, fixing problems for hundreds of checks at a time instead of tediously resolving them one by one. In cases where legacy tools would force teams to slog through, for example, 3,400 checks with 600 million errors, Vision AI’s

clustering can reduce that effort to investigating just 381 groups—turning mountains into molehills and speeding debug time by at least 2x.

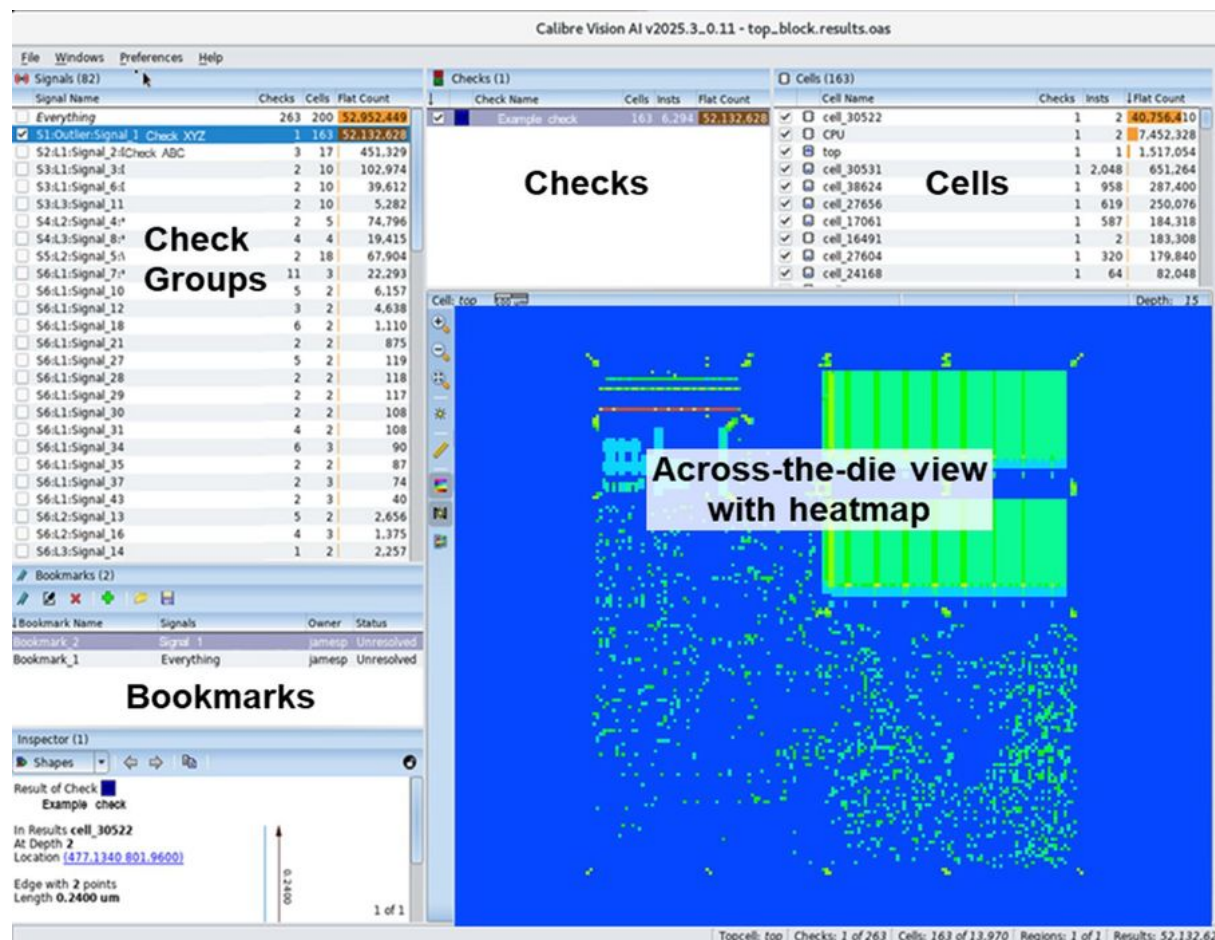


Figure 2. The Calibre Vision AI software automates and simplifies the chip-level DRC verification process.

Vision AI is also highly collaborative. Dynamic bookmarks capture the exact state of analysis, from layer filters to zoomed layout areas, along with annotations and owner assignments. Sharing a bookmark sends a living analysis—not just a static snapshot—to coworkers, so everyone is working from the same view. Teams can export results databases, distribute actionable groups to block owners, and seamlessly import findings into other Siemens EDA tools for further debug.

Empowering every designer: reducing the expertise gap

A frequent pain point in chip verification is the need for deep expertise—knowing which errors matter, which patterns mean trouble, and how to interpret complex results. Calibre Vision AI helps level the playing field. Its AI-based algorithms consistently create the same clusters and debug paths that senior experts would identify, but does so in minutes. New users can quickly find systematic issues and perform like seasoned engineers, helping chip companies address workforce shortages and staff turnover.

Beyond clusters and bookmarks, Vision AI lets designers build custom signals by leveraging their own data. The platform secures customer models and data for exclusive use, making sure sensitive information stays within the company. And by integrating with Siemens' EDA AI ecosystem, Calibre Vision AI supports generative AI chatbots and reasoning assistants. Designers can ask direct questions—about syntax, about a signal, about the flow—and get prompt—accurate answers, streamlining training and adoption.

Real results: speeding analysis and sharing insight

Customer feedback from leading IC companies shows the real-world value of AI for full-chip DRC analysis and debug. One company reported that Vision AI reduced their debug effort by at least half—a gain that makes the difference between tapeout and delay. Another noted the platform's signals algorithm automatically creates the same check groups that experienced users would manually identify, saving not just time but energy.

Quantitative gains are dramatic. For example, Calibre Vision AI can load and visualize error files significantly faster than traditional debug flows. Figure 3 shows the difference in four different test cases: a results file that took 350 minutes with the traditional flow, took Calibre Vision AI only 31 minutes. In another test case (not shown), it took just five minutes to analyze and cluster 3.2 billion errors from more than 380 rule checks into 17 meaningful groups. Instead of getting lost in gigabytes of error data, designers now spend time solving real problems.

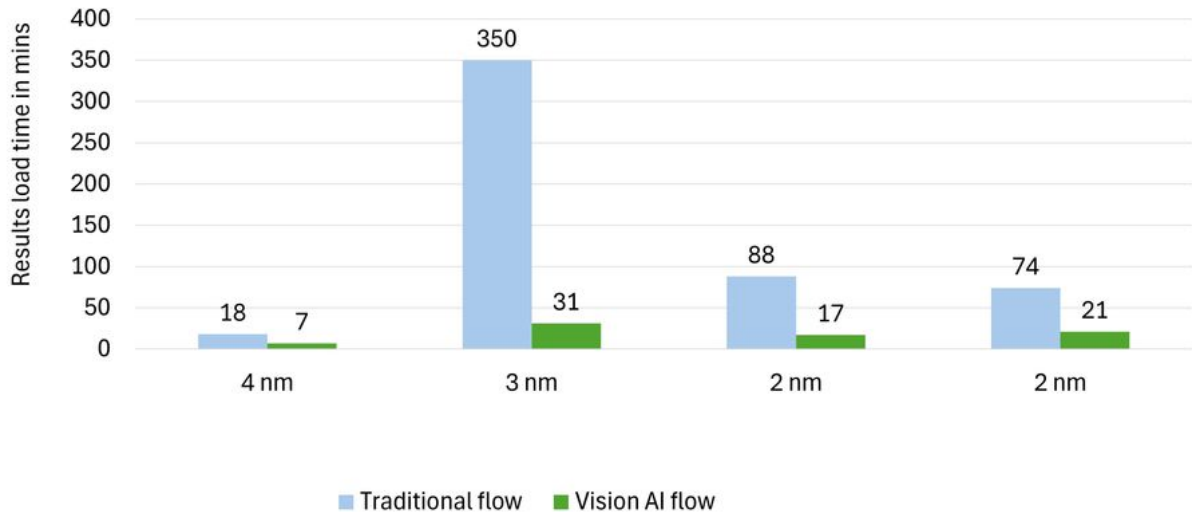


Figure 3. Charting the results load time between the traditional DRC debug flow and the Calibre Vision AI flow.

Looking ahead: the future of AI in chip design

Today's chips demand more than incremental improvements in EDA software. As the need for speed, quality and collaboration continues to grow, the story of physical verification will be shaped by smarter, more adaptive technologies. With AI-powered DRC analysis, we see a clear path: a faster and more productive way to find systematic issues, intelligent debug, stronger collaboration and the chance for every designer to make an expert impact.

By combining the creativity of engineers with the speed and insight of AI, platforms like Calibre Vision AI are driving a new productivity curve in full-chip analysis. With these tools, teams don't just keep up with complexity—they turn it into a competitive advantage.

At Siemens, the future of chip verification is already taking shape—where intelligence works hand in hand with intuition, and new ideas find their way to silicon faster than ever before. As the industry continues to push boundaries and unlock the next generation of devices, AI will help chip design reach new heights.